# Revisiting Dalgado: Tracing the Heritage of the Portuguese Language in South Asia

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## Introduction

The current submission describes some of the latest developments in the Cultural HeritAge and Multilingual Understanding through lexiCal Archives **(CHAMUÇA)** initiative that seeks to create a (linked data) knowledge graph describing the impact of the Portuguese language on the vocabularies of numerous Asian languages. CHAMUÇA is also the name of the resulting (open) resource which will be published in multiple versions. This resource combines both pre-existing resources (curated and checked by experts) and original scholarship. Indeed, one of the central aims of CHAMUÇA is to make past and existing scholarship in this area (often dispersed and/or unavailable electronically or online) more accessible both to researchers and the general public in a highly machine-actionable form. It does this, in part, by making its results available as linked open data, via a Wikibase instance, and, more generally, by following the FAIR principles in the creation and publication of this data. In what follows, we look at CHAMUÇA-HiUr , the first version of the CHAMUÇA knowledge graph which covers Portuguese borrowings in the Urdu ( *ur* ) and Hindi ( *hi* ) languages. Moreover, we describe how we went about integrating different kinds of historical, cultural, and linguistic information in creating this first version of the resource.

Indeed, we made extensive (though not exclusive) use of two different kinds of sources in building this version of CHAMUÇA. The first were open wiki-based resources such as Wiktionary, Wikidata and Wikipedia from which we extracted a large amount of basic linguistic information. The second kind were legacy resources which detailed previous scholarship in this area; in particular, we sought to incorporate (and update) the groundbreaking research of Sebastião Rodolfo Dalgado, a Catholic priest and academic active in the latter half of the 19th and beginning of the 20th centuries, into our resource. In 1913, Dalgado published *Influencia do Vocabulário Português em Línguas Asiáticas* (Dalgado, 1913), which remains one of the definitive works on the topic of Portuguese loanwords in Asian languages to this day; an English translation of the original Portuguese text was published a couple of decades latter as *Influence of Portuguese Vocables in Asiatic Languages* (Soares, 1936). In addition to a lengthy expository introduction, (Dalgado, 1913) contains a lexicon consisting of entries with Portuguese headwords followed by lists of words in Asian languages deriving from that original etymon. We have kept to this mode of organisation in CHAMUÇA, using a separate Portuguese lan-

guage lexicon to index the other lexicons in the resource, Fig. 1 below shows the architecture of CHAMUÇA-HiUr. Further details about the organisation of CHAMUÇA can be found in (Khan et al., (2024a)) (Khan et al., (2024b)).

Dalgado as well as being a native speaker of Konkani, the Indo-Aryan language spoken in Goa, was proficient in a number of South Asian languages as well as Indonesian (along with being fluent in Portuguese of course). The languages covered in (Dalgado, 1913) are therefore mostly those spoken in South Asia (e.g., Hindi-Urdu, Konkani, Tamil, Bengali, and Malayalam) although other Asian languages such as Japanese are also covered to a certain extent. Similarly, give the respective linguistic and research competences of the authors of the current paper, the first few releases of CHAMUÇA will also focus on South Asian languages. In the next section we look at the case of Portuguese borrowings into Hindi and Urdu.

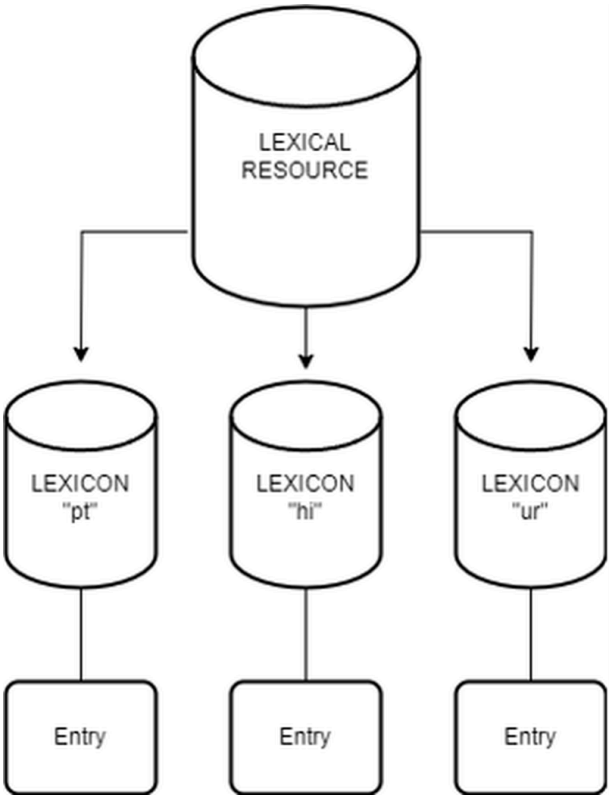


Fig. 1 Organisation of CHAMUÇA-HiUr

## Case Study: The Influence of Portuguese on Hindi and Urdu

Our first release (which is downloadable from https://github.com/anasfkhan81/chamuca/tree/main/chamuca_lex_resource ), **CHAMUÇA-HiUr** , is as mentioned, a lexicon of Portuguese borrowings into Hindi-Urdu[1] . We chose these two languages to start with both for practical reasons (including the ready availability of both native language experts and accessible resources which we could consult) but also because we felt it would be interesting to start with languages which weren't associated with regions directly affected by the arrival of Portuguese settlers and colonists -- but which ended up being impacted by Portuguese nonetheless[2] . Our starting point in the creation of CHAMUÇA-HiUr was Dalgado's list of Hindi and Urdu words borrowed from Portuguese, these number 107 in Urdu and 53 in Hindi (with almost all of the Hindi words shared with Urdu). We cross-checked this list with a number of other lexicographic resources – in particular, we consulted the McGregor (McGregor, 1993) and Platts' dictionaries (Platts, 1884) both of which have been made available on the excellent Digital Dictionaries of South Asia Portal[3] -- and as a result we added a number of words which weren't in Dalgado's original. Both Hindi and Urdu contain more or less the same loanwords from Portuguese (because of the historical period in which almost all of these words entered the lexicons of both languages), and so we increased the number of words in Hindi by including those Urdu words for which we found citations in other resources. Since Dalgado's entries were written exclusively using the Roman alphabet (and feature transliterations of the loan words in question using Dalgado's own system of transliterations) we had to find the corresponding words in Hindi (Devanagari) and Urdu (modified Arabic script) in other lexicographic resources – something which wasn't always straightforward. In addition to including linguistic information derived from other resources, we also incorporated a number of etymologies not in Dalgado. We also added corpus frequency information from two modern corpora: **Hindi Web 2021** (hiTenTen21) and **Urdu Web 2018** (urTenTen18), for Hindi and Urdu respectively (Jakubíček et al. 2013) in order to reflect the contemporary usage of such words. The whole lexical resource consisting of just over a hundred words, which in the vast majority of cases are shared by both languages, can be downloaded from our GitHub page[4] : this will shortly also be deposited in the ILC4CLARIN[5] centre. We also plan to publish CHAMUÇA-HiUr, via Wikibase and via a linguistic knowledge graph which we are currently putting together as part of the upscaling of the CLARIN-IT node funded by the H2IOSC[6] project.

Acknowledgements

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# Notes

1. In the case of Urdu, Dalgado uses ‘Hindustani’ to refer to the more colloquial everyday language and ‘Urdu’ to refer to the literary register but this distinction isn’t maintained nowadays so we do not use it.
2. Anecdotally our experience has been that speakers of these two languages don’t tend to be aware of the impact that Portuguese has had on either Hindi or Urdu (in contrast to speakers of Konkani or Sinhalese for instance, again in our experience) despite the fact that a number of very common words, such as those for key or room or bucket, derive from that language.
3. https://dsal.uchicago.edu/dictionaries/
4. https://github.com/anasfkhan81/chamuca/tree/main/chamuca_lex_resour ce
5. https://ilc4clarin.ilc.cnr.it/
6. https://www.h2iosc.cnr.it/

# Bibliography

**Dalgado, S. R. (1913):** *Influência do vocabulário português em línguas asiáticas (abrangendo cerca de cinquenta idiomas)* **.** Coimbra: Imprensa da Universidade.

**Khan, Fahad / Salgado, Ana / Anuradha, Isuri / Costa, Rute / Liyanage, Chamila / McCrae, John Philip / Ojha, Atul Kumar / Rani, Priya / Frontini, Francesca (2024a):** *CULTURAL HERITAGE AND MULTILINGUAL UNDERSTANDING THROUGH LEXICAL ARCHIVES (CHAMUÇA).* Euralex 2024.

**Khan, Fahad / Salgado, Ana / Anuradha, Isuri / Costa, Rute / Liyanage, Chamila / McCrae, John Philip / Ojha, Atul Kumar / Rani, Priya / Frontini, Francesca (2024b):** *“CHAMUÇA: Towards a Linked Data Language Resource of Portuguese Borrowings in Asian Languages”,* in: Proceedings of the 9th Workshop on Linked Data in Linguistics @ LREC-COLING 2024: 44–48.

**Jakubíček, Miloš / Kilgarriff, Adam / Kovář, Vít / Rychlý, Pavel / Suchomel, Vít (2013):** *The TenTen corpus family,* in: 7th International Corpus Linguistics Conference CL: 125–127.

**McGregor, R. S. (1993):** *The Oxford Hindi-English Dictionary* **.** New York: Oxford University Press. <https://dsal.uchicago.edu/dictionaries/mcgregor> [30.04.2025]

**Platts, John T. (1884):** *A dictionary of Urdu, classical Hindi, and English.* London: W. H. Allen & Co. <https://dsal.uchicago.edu/dictionaries/platts/> [30.04.2025]

**Soares, A. X. (1936):** *Portuguese Vocables in Asiatic Languages from the Portuguese original of Monsignor Sebastião Rodolfo Dalgado* , trans. with notes, additions, comments . Baroda: p. XXXIII, pp. 313–314.